Home

EDA dashboard

ML Models

📊 EDA Dashboard

Air Quality · 13 Tabs

s ★ 8 · Weather vs Pollution ★ 9 · A/B Test ★ 10 · Time Series Decom

💡 Tab 13 — Insights & Recommendations

🌍 Air Quality Analysis — Final Report

Dataset: UCI Air Quality · Italy · March 2004 – April 2005 · 9,357 hourly records

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1 Data Quality

✅ NMHC(GT) dropped — 90.3% missing, structurally incomplete sensor.

✅ CO(GT), NOx(GT), NO2(GT) had ~19% missing — successfully imputed with median.

✅ -200 null marker correctly identified and replaced before all analysis.

✅ After cleaning: dataset is complete, stationary, and ML-ready.

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2 Pollution Drivers

✅ Traffic is the primary CO source — confirmed by AM/PM rush hour peaks (+30–40% above baseline).

✅ Weekdays have 20–30% higher CO than weekends — business traffic dominates urban pollution.

✅ Winter months show highest pollution: cold air thermal inversion + heating combustion + reduced dispersion.

✅ Benzene (C_6H_6) tracks CO closely — both are vehicle exhaust combustion products.

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3 Weather Interactions

✅ Cold temperature → higher pollution. Thermal inversion traps pollutants in the boundary layer.

✅ Higher humidity mildly reduces CO — wet conditions help particle washout and gas absorption.

⚠️ Temperature alone explains ~20% of CO variance — weather is important but not the only factor.

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4 Sensor Reliability

✅ Metal-oxide sensors (PT08.Sx) correlate $r > 0.85$ with reference analyzers — reliable for monitoring.

✅ Low-cost sensors can serve as early-warning systems, but reference analyzers needed for compliance.

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5 Statistical Findings

✅ A/B Test confirmed: rush hour CO is significantly higher than off-peak ($p < 0.05$, Cohen's d medium-large).

✅ All GT pollutants are stationary — mean-reverting, no permanent drift over the study period.

✅ Distributions are right-skewed — pollution is low most of the time, with rare severe events.

6️⃣ Recommendations

🚦 **Traffic Policy:** Implement odd-even or congestion pricing during rush hours (7–9h, 17–19h) to directly cut CO.

❄️ **Winter Alert System:** Deploy real-time CO alerts during cold spells — thermal inversion creates dangerous hotspots.

🔬 **Sensor Network:** Low-cost PT08 sensors are reliable enough for a dense city-wide monitoring network at low cost.

📊 **Forecasting:** Use ML models (trained in Stage 2) to predict next-hour CO — enable preemptive public health warnings.

🌱 **Green Infrastructure:** Vegetation buffers on major roads reduce pollutant dispersion path — supported by this data's spatial pattern.

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